SciPub course, seminar 4, Research Methods & Validity

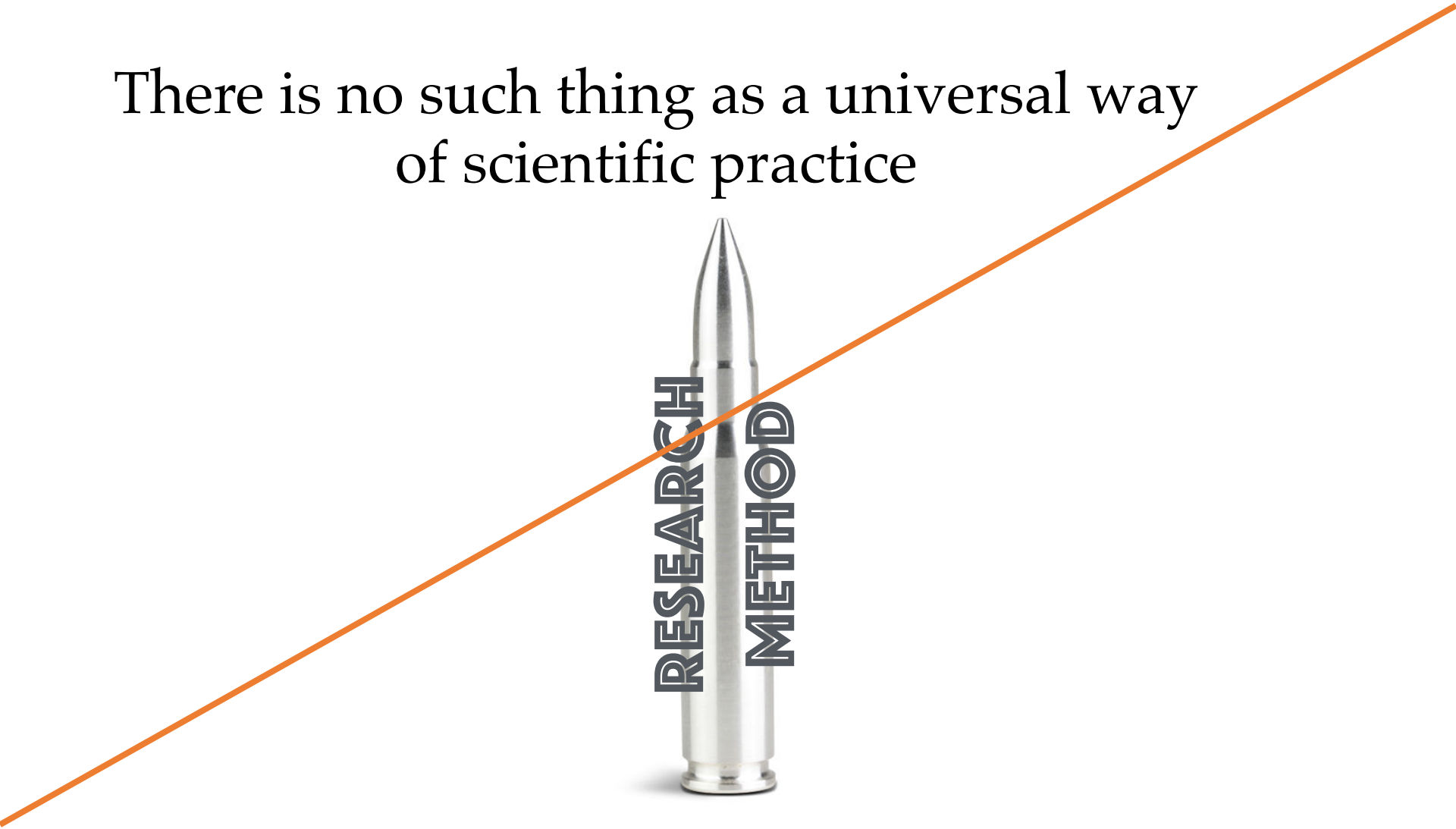
20261104, NTNU, Trondheim

Robert Feldt



Which research method(s) to use in which situation?

There is no such thing as a universal way of scientific practice



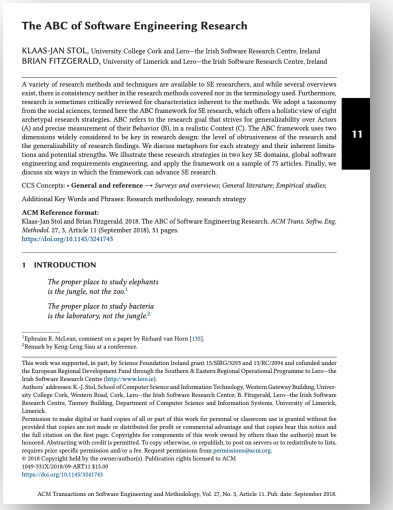
Method selection depends on many non-trivial questions

- What is the purpose of the study?
  Exploratory? Descriptive? Explanatory? Improving?
- What is the nature of the study?
  Inductive? Deductive?
- What is the relation to existing evidence?
  Building a new theory? "Testing" existing theory?
- What is the nature of the questions we ask?
  What-questions? Why-questions?
- What is the nature of the environment?
  Controlled environments? Realistic environments?
- What is the necessary sample?
  Population source?
  Units of analysis?

Criteria for selecting methods

Criteria for environment selection (and sampling)

Not trivial, but possible: checklists



The ABC of Software Engineering Research

KLAAS-JAN STOL, University College Cork and Lero—the Irish Software Research Centre, Ireland
BRIAN FITZGERALD, University of Limerick and Lero—the Irish Software Research Centre, Ireland

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A variety of research methods and techniques are available to SE researchers, and while several overviews exist, there is consistency neither in the research methods covered nor in the terminology used. Furthermore, research is sometimes critically reviewed for characteristics inherent to the methods. We adopt a taxonomy from the social sciences, termed here the ABC framework for SE research, which offers a holistic view of eight archetypal research strategies. ABC refers to the research goal that strives for generalizability over Actors (A) and precise measurement of their Behavior (B), in a realistic Context (C). The ABC framework uses two dimensions widely considered to be key in research design: the level of obtrusiveness of the research and the generalizability of research findings. We discuss metaphors for each strategy and their inherent limitations and potential strengths. We illustrate these research strategies in two key SE domains, global software engineering and requirements engineering, and apply the framework on a sample of 75 articles. Finally, we discuss six ways in which the framework can advance SE research.

CCS Concepts: • **General and reference** → *Surveys and overviews; General literature; Empirical studies;*

Additional Key Words and Phrases: Research methodology, research strategy

ACM Reference format:
Klaas-Jan Stol and Brian Fitzgerald. 2018. The ABC of Software Engineering Research. *ACM Trans. Softw. Eng. Methodol.* 27, 3, Article 11 (September 2018), 51 pages.
https://doi.org/10.1145/3241743

1 INTRODUCTION

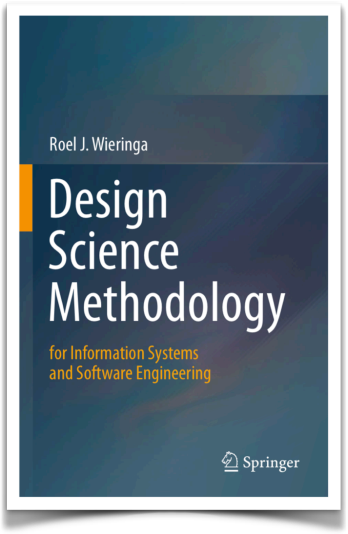
The proper place to study elephants is the jungle, not the zoo.[1]

The proper place to study bacteria is the laboratory, not the jungle.[2]

[1]Ephraim R. McLean, comment on a paper by Richard van Horn [135].
[2]Remark by Keng-Leng Siau at a conference.

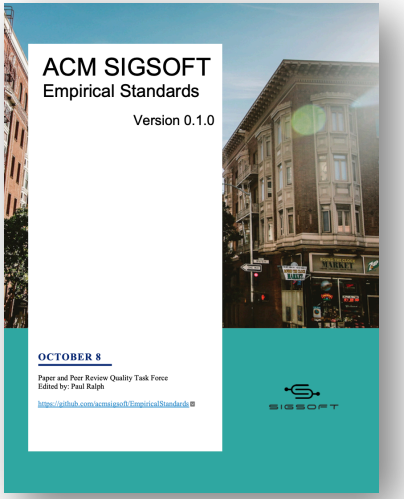
This work was supported, in part, by Science Foundation Ireland grant 15/SIRG/3293 and 13/RC/2094 and cofunded under the European Regional Development Fund through the Southern & Eastern Regional Operational Programme to Lero—the Irish Software Research Centre (http://www.lero.ie).
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Good starting point on positioning of research methods

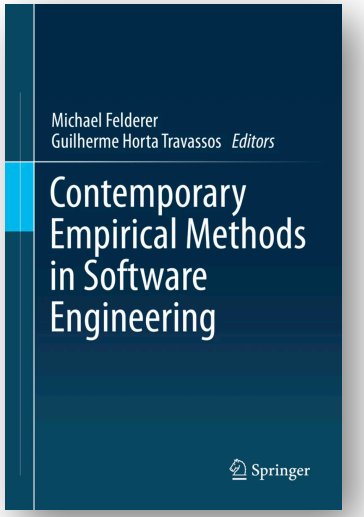


More advanced

- Chapter 16 + Appendix
- http://bit.ly/checklists-design_science



Overview of quality criteria as per recommendation by research community:



Deep dives into selected research methods

The ABC of Software Engineering Research

KLAAS-JAN STOL, University College Cork and Lero—the Irish Software Research Centre, Ireland
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Table 2. Examples of Actors, Behavior, and Context in Software Engineering Research

Actors	Managers, software engineers, users, software systems, software development artifacts incl. defects, tools, techniques, prototypes
Behavior	System behavior (e.g., reliability, performance, and other quality attributes), software engineers' behavior and antecedents such as productivity, motivation, and intention
Context	Industrial settings, organizations, software projects, development teams, software laboratory, classroom, meeting rooms

Table 3. Comparison of Knowledge-Seeking versus Solution-Seeking Research in Software Engineering Research

	Knowledge-Seeking Research	Solution-Seeking Research
Goal	To generate or propose scientific claims and to evaluate and validate those claims. This may also include the development of instruments or other artifacts with the specific purpose of supporting or enabling these knowledge-seeking activities, e.g., the construction of an instrument that facilitates data gathering or analysis.	To design or develop new or improve existing solutions that can help to overcome or ameliorate challenges, bottlenecks, and other problems in the development of software systems and supporting processes.
Focus of research	A phenomenon within software engineering, or a characteristic thereof, that is not (or not sufficiently) well understood or known.	A specific software engineering challenge, obstacle, or problem, and the design or creation of a solution.
Outcome	Empirical findings, descriptions, insights generated by simulations, theoretical or conceptual frameworks, or hypotheses.	Artifacts that include algorithms, tools, notations (incl. languages), models, mechanisms, and techniques.
Example	Research question: How does Extreme Programming work? Sharp et al. conducted an ethnographic study to capture how developers employ XP development practices [180].	Research question: how to select a manageable subset of the input data faster in order to automatically find performance bottlenecks? Luo et al. proposed FOREPOST: an adaptive, feedback-directed learning testing system that helps to find performance bottlenecks [124]. (Note that the *evaluation* of FOREPOST is a knowledge-seeking study.)

Degree of "manipulation"

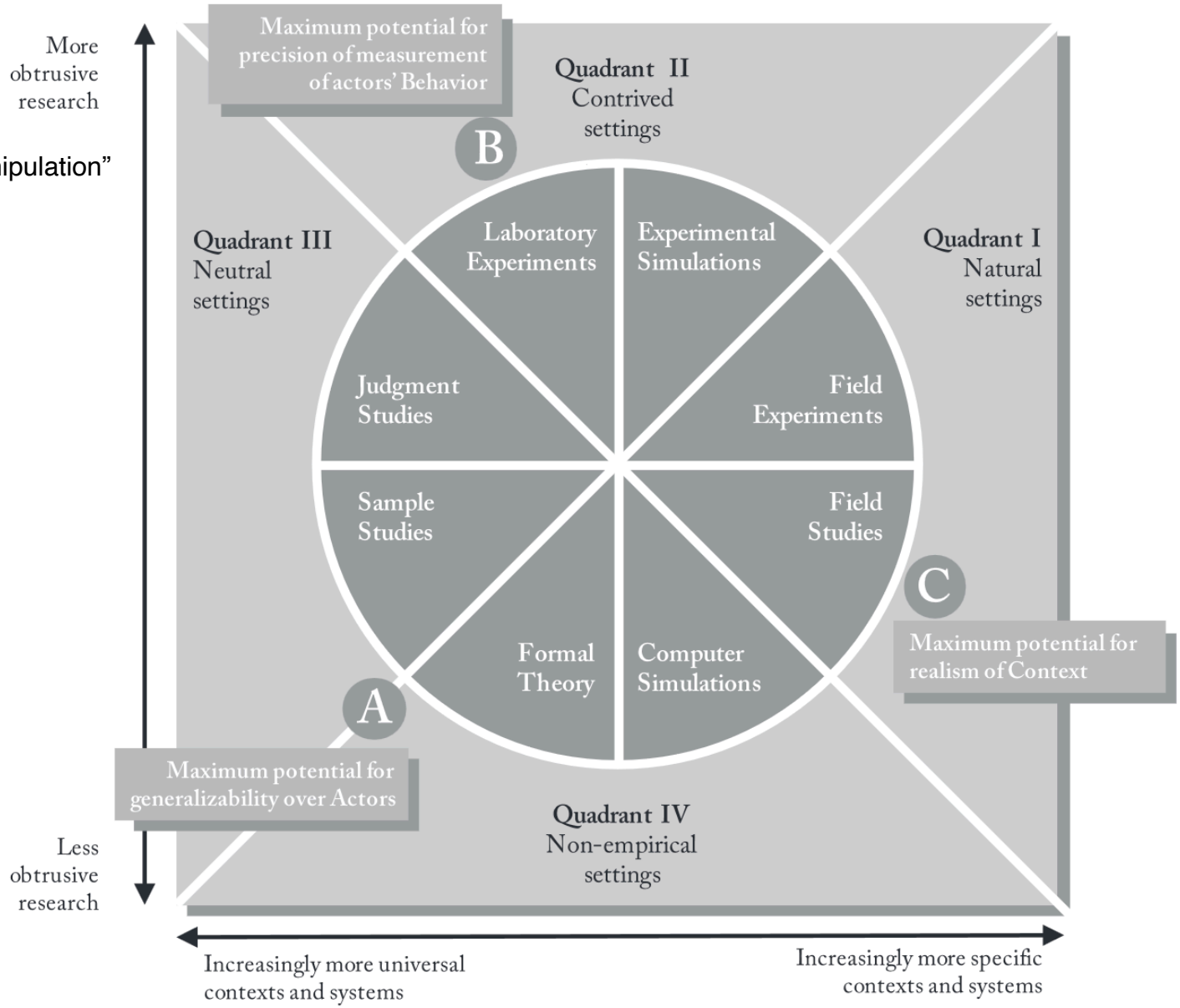


Fig. 1. The ABC framework: eight research strategies as categories of research methods for software engineering (adapted from Runkel and McGrath [170]). Quadrants I to IV represent different research settings.

Who, what, how: do we study the people we claim to help?

The who, what, how of software engineering research: a socio-technical framework

Check for updates

2020

Margaret-Anne Storey[1] · Neil A. Ernst[1] · Courtney Williams[1] · Eirini Kalliamvakou[1]

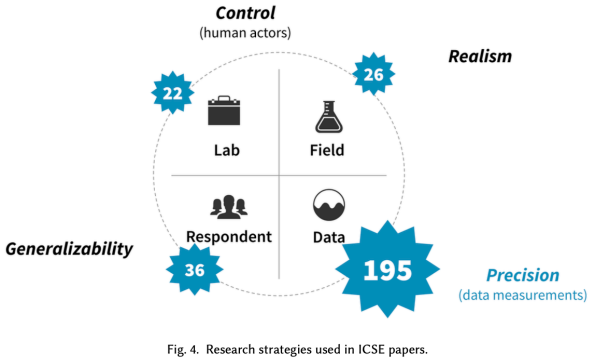


Fig. 4. Research strategies used in ICSE papers.

- 3 questions for any study:
  - **Who** benefits? (humans, systems, researchers)?
  - **What** is the contribution? (Descriptive, solution, …)
  - **How** is it studied? (Field, lab, respondent, or data strategies)
- 54% of ICSE papers studied and 43% of EMSE studies:
  - Claims to benefit humans but never study humans directly
- Only 25% triangulate across strategy quadrants
- Among ICSE authors:
  - 17 of 60 authors didn't know what triangulation means
- **Punchline:** use Who-What-How when **designing** your study. If your "who" is people, your "how" should include them.

Mixing methods with a reason

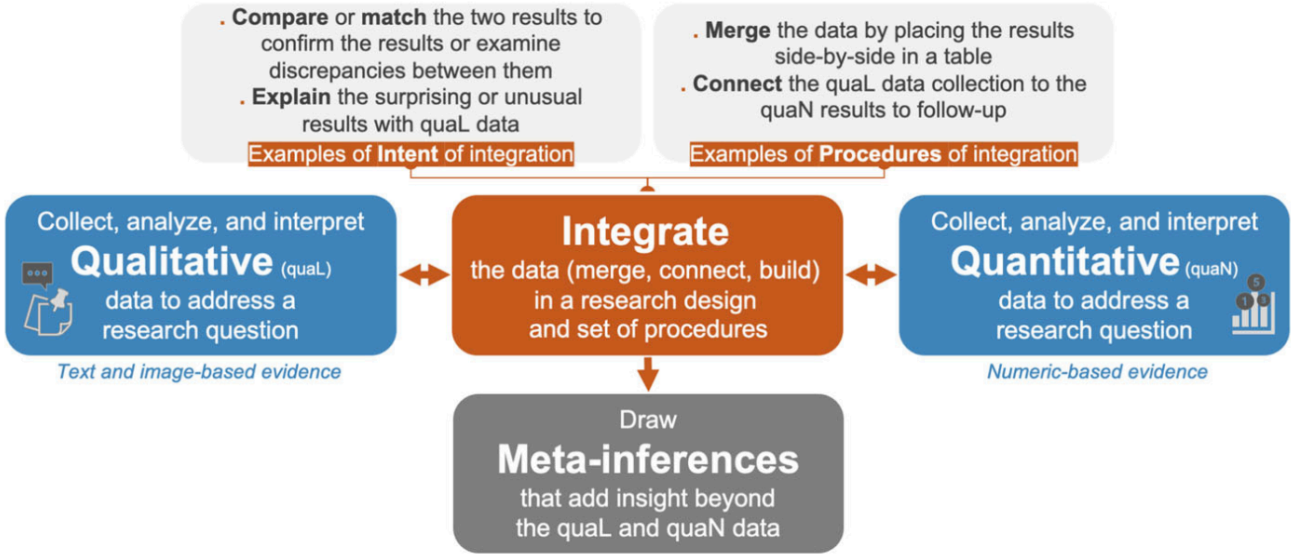
Empirical Software Engineering (2025) 30:138
https://doi.org/10.1007/s10664-025-10629-x

Check for updates

Guiding principles for mixed methods research in software engineering

Margaret-Anne Storey[1] · Rashina Hoda[2] · Alessandra Maciel Paz Milani[1] · Maria Teresa Baldassarre[3]

- Mixed methods = collect, analyse and **integrate** qualitative and quantitative data to produce insights neither gives alone



Mixing methods with a reason

Check for updates

Guiding principles for mixed methods research in software engineering

Margaret-Anne Storey[1] · Rashina Hoda[2] · Alessandra Maciel Paz Milani[1] · Maria Teresa Baldassarre[3]

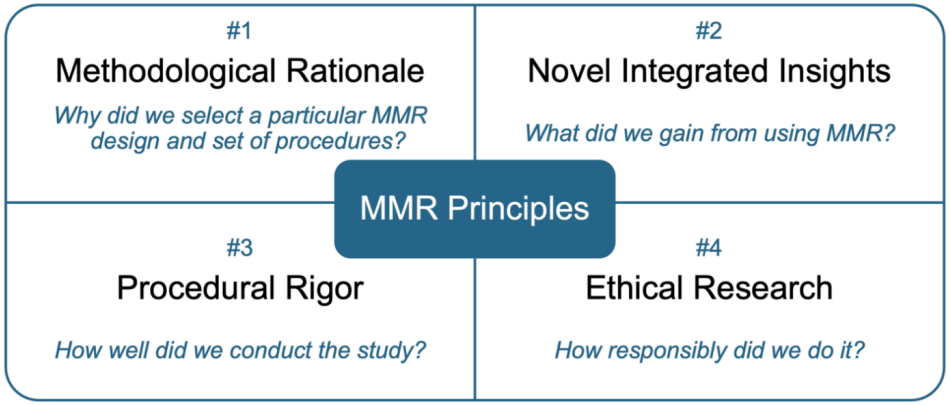


Fig. 2 Four guiding principles for mixed method research in software engineering

- Four MMR designs:
  - exploratory sequential (qual → quant, e.g. interviews → survey);
  - explanatory sequential (quant → qual, e.g. mining → interviews to explain);
  - convergent parallel (both at once, compared);
  - embedded (e.g. a survey inside an experiment).
- Many papers mix methods without motivating the design

Mixing methods with a reason

Guiding principles for mixed methods research in software engineering

Margaret-Anne Storey[1] · Rashina Hoda[2] · Alessandra Maciel Paz Milani[1] · Maria Teresa Baldassarre[3]

- Antipatterns
  - **selling your soul** = method added only to please reviewers
  - **smoke and mirrors** = token second strand
  - **uninvited guest** = a method first appears in the discussion
  - **integration failure** = collects qualitative and quantitative data but never meaningfully combines the findings
    - Check: "Does your discussion section contain a conclusion that needs both kinds of data to support it?"
  - **lost opportunity =** in a sequential design, the findings from the first strand don't shape the instruments for the next one (e.g. the interviews didn't inform the survey questions).

Threats to Validity in Software Engineering – hypocritical paper section or essential analysis?

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Threats to validity: essential analysis or ritual?

- Analysed 91 ICSE papers with ACM SIGSOFT Distinguished Paper Awards
- Threats seldom discussed in depth
  - mostly an **enforced afterthought**, not part of design and execution.
- 12 items to consider, for researchers, reviewers and readers:
  - When designing:
    - decide which validity types matter most for your goal and focus there;
    - reason about trade-offs between threats;
    - use categories that fit your study type (most papers don't: only 32% of the 91 do).
  - When reporting:
    - analyse in depth, covering priorities, mitigations, rationale and what remains unmitigated;
    - write for the users of the research, not for the reviewers.
  - When reviewing:
    - judge validity against the paper's claims, not by whether a section exists;
    - do not punish honesty about threats and limitations.
- When reading: interpret and cite results in light of their validity.

Don’t list threats - explain your trade-offs!

Communicating Study Design Trade-offs in Software Engineering

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Communicating Study Trade-offs in SE

- **Problems** with today's threats sections:
  - boilerplate: "sample size limits external validity" is a truism
  - defensive writing: choices are played down to pre-empt reviewers
  - opacity: the reader sees the limitation but can't judge whether the choice was reasonable
- Robillard et al propose:
  - every design decision is one option among alternatives
  - report why its benefits outweigh the costs of those alternatives, and
  - let threats appear as consequences of decisions.
- Five parts for each major trade-off:
  - **Decision point**: what you decided.
  - **Alternatives**: what else you could have done.
  - **Considerations**: the costs, benefits and constraints of each, with evidence where possible.
  - **Rationale**: why this option.
  - **Implications**: the consequences for your findings, including threats.

Your analysis choices can flip your result

Exploring the Garden of Forking Paths in Empirical Software Engineering Research: A Multiverse Analysis

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Your analysis choices can flip your result

- **Multiverse re-analysis** of a published MSR study
- Identified 9 key decision points and defensible alternatives for them
- Re-ran 3,072 different analyses
- Only 6 of them (0.2%) reproduced all of the published findings
- Many gave opposite claims
- Biggest effect comes from period length, a decision usually made by default or convention

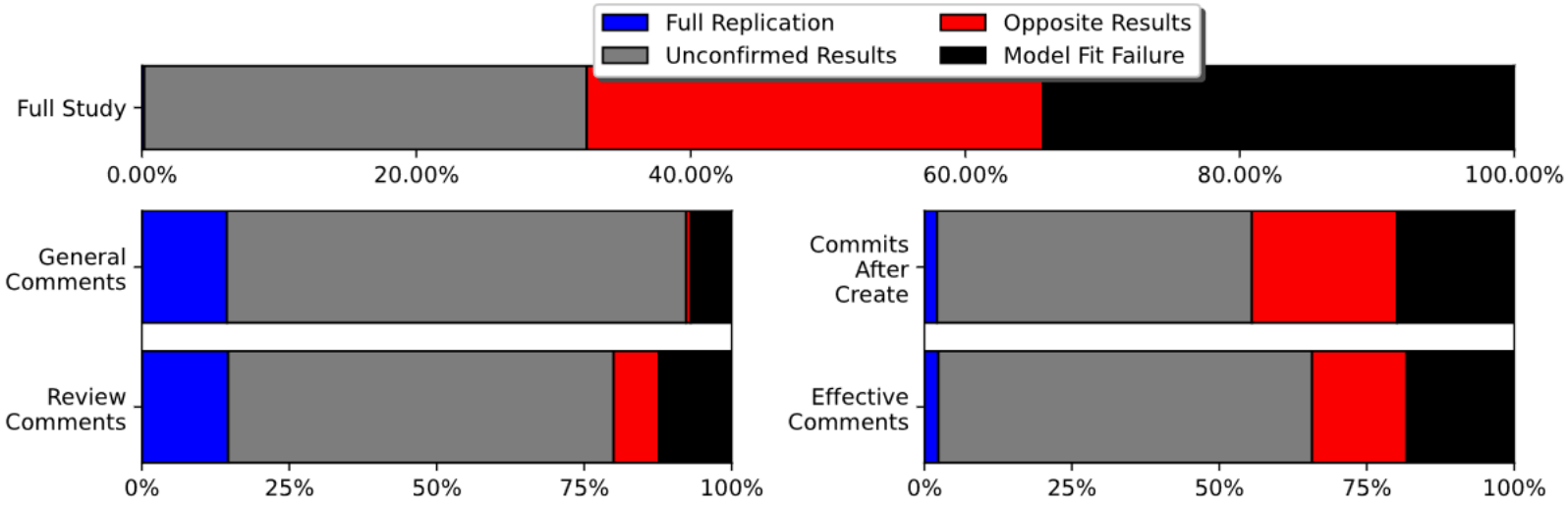


Fig. 3. High-level overview of the outcomes for all of the multiverses, per dependent variable.

Table 5. The JLAC with Stronger and More Preferable Levels of Justification on Higher Levels (5) of the Ladder

Level	Name	Description
5	Causal	These decisions are made based on causal models or theories of the problem, or domain, under study.
4	Empirical	These decisions are justified based on properties of the data under study, but these decisions are not necessarily linked to a causal model.
3	Heuristic	These are decisions made based on general heuristics, often made based on reasoning that it is not directly tied to the domain or the RQ being studied.
2	Conventional	These decisions are made based on existing conventions, where the justification of the decision is made because it is common practice in the field.
1	Implicit	These are decisions that are made implicitly, without the researcher being aware the decisions are being made. These could be defaults of the statistics package being used.

Your research methods?

Let's interact & discuss!

- 0. Think for yourself for 5 minutes, which research methods do I typically use? And Why?
- 1. Team up in pairs/triples
- 2. Each person presents their study in 2 minutes & partner(s) plays reviewers:
  - Which threat matters most for your main claim, and what did you trade away?
- 3. Consider if any of these questions applies to or could help you refine your study:
  - **Who** is meant to benefit from your study, and does your **how** actually include them?
  - Could an **alternative method** be a second strand, and which of the four MMR designs would it be?
  - Are there alternative decisions you can make in your analysis?
- 4. After circa 15-20 minutes we discuss back in full class